

MUTUAL FUND ANALYTICS PLATFORM

Bluestock Fintech Capstone | End to End Data Engineering & Analytics

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Data Scale: 46,000+ NAV Records

Institution: MIT Manipal
AMCs Anchored: SBI MF, HDFC, ICICI, Nippon (10 Total)

KEY INDUSTRY BENCHMARKS

INR 31,002 Cr Dec 2025 Peak SIP Inflow	INR 12.5L Cr SBI MF AUM Anchor	26.12 Cr Industry Folio Count	40 Schemes AMFI Schemes Tracked
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STAR SCHEMA ARCHITECTURE & ANALYTICS

This capstone platform ingests public Indian mutual fund data from AMFI and mfapi.in, cleans and loads raw NAV history, quarterly AUM, and investor transactions into a normalized 5+ table Star Schema (dim_fund, fact_nav, fact_aum, fact_sip, fact_transactions, dim_investor).

- ETL Script: Automated Python pipeline in scripts/etl_pipeline.py loading into SQLite via SQLAlchemy
- Advanced Risk Analytics: Computes Sharpe Ratio, Sortino Ratio, VaR (95%), Max Drawdown, Jensen Alpha, and Beta against Nifty 50
- Demographic Insights: Analyzed 32,000+ transactions across 5,000 investors in 12 Indian states
- Interactive Dashboard: 4-page Streamlit analytics web application

DATASET SUMMARY (10 REAL WORLD FILES)

File Name	Record Count	Description
01_fund_master.csv	40	40 AMFI schemes across 10 AMCs with expense ratio, manager, and risk grade
02_nav_history.csv	46,000+	Daily NAV history from Jan 2022 to May 2026
03_aum_by_fund_house.csv	90	Quarterly AUM for top 10 AMCs (2022 to 2025)
04_monthly_sip_inflows.csv	48	Monthly SIP inflows peaking at INR 31,002 Cr milestone
08_investor_transactions.csv	32,000+	Transactions across 12 Indian states with age, income, and city tier